$\left[\text{K.E}\left(\text{r}=\infty\right)=\frac{1}{2}\text{Mv}_{\infty}{}^{2}\right]$, with what velocity should the object be thrown from Earth?

Ans: $v_e=\sqrt{v_\infty^2+2gR_E}$

14. Suppose we go 200 km above and below the surface of the Earth, what are the g values at these two points? In which case, is the value of g small?

Ans: $g_{down} = 0.96\ g$

$g_{up} = 0.94\ g$

15. Calculate the change in g value in your district of Tamil nadu. (Hint: Get the latitude of your district of Tamil nadu from the Google). What is the difference in g values at Chennai and Kanyakumari?

Ans: $g_{chennai} = 9.7677\ m\ s^{-2}$

$g_{Kanyakumari} = 9.7667\ m\ s^{-2}$

$\Delta g = 0.001\ m\ s^{-2}$

V. Conceptual Questions

1. In the following, what are the quantities which that are conserved?

a) Linear momentum of planet

b) Angular momentum of planet

c) Total energy of planet

d) Potential energy of a planet

2. The work done by Sun on Earth in one year will be

a) Zero b) Non zero

c) positive d) negative

3. The work done by Sun on Earth at any finite interval of time is

a) positive, negative or zero

b) Strictly positive

c) Strictly negative

d) It is always zero

4. If a comet suddenly hits the Moon and imparts energy which is more than the total energy of the Moon, what will happen?

5. If the Earth's pull on the Moon suddenly disappears, what will happen to the Moon?

6. If the Earth has no tilt, what happens to the seasons of the Earth?

7. A student was asked a question 'why are there summer and winter for us? He replied as 'since Earth is orbiting in an elliptical orbit, when the Earth is very far away from the Sun(aphelion) there will be winter, when the Earth is nearer to the Sun(perihelion) there will be winter'. Is this answer correct? If not, what is the correct explanation for the occurrence of summer and winter?

8. The following photographs are taken from the recent lunar eclipse which occurred on January 31, 2018. Is it possible to prove that Earth is a sphere from these photographs?



UNIT-6(XI-Physics_Vol-2).indd 46 02-01-2023 14:30:05